

VMF Coverity Analysis



Visteon Innovation & Technology GmbH

Mai 2016

Installation of local Coverity

Download and install

<http://ci.visteon.com/raw/slave-win/installs/cov-analysis-win64-7.5.0.exe>

or from NAS \Tools\Coverity\cov-analysis-win64-7.5.0.exe

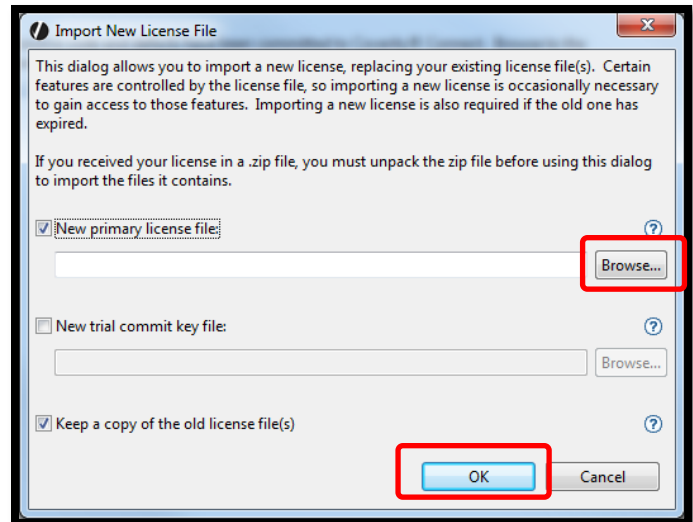
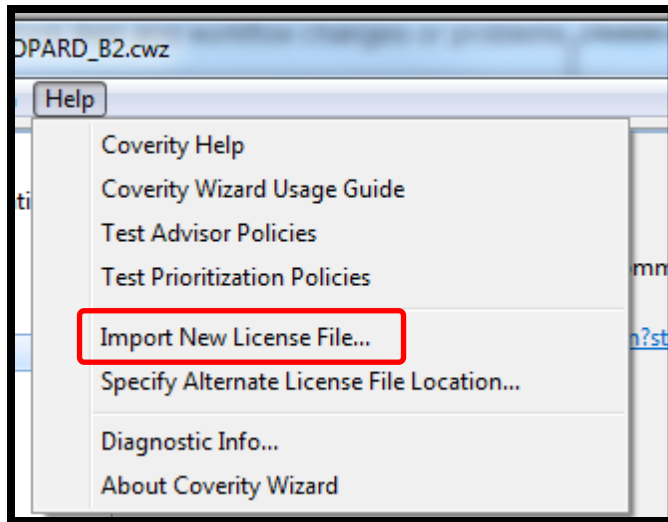
Get Coverity license form ci.visteon.com

<http://ci.visteon.com/raw/coverity-global-host-license.dat>

Install Coverity license

Start **Coverity Wizard** from Windows Start Menu.

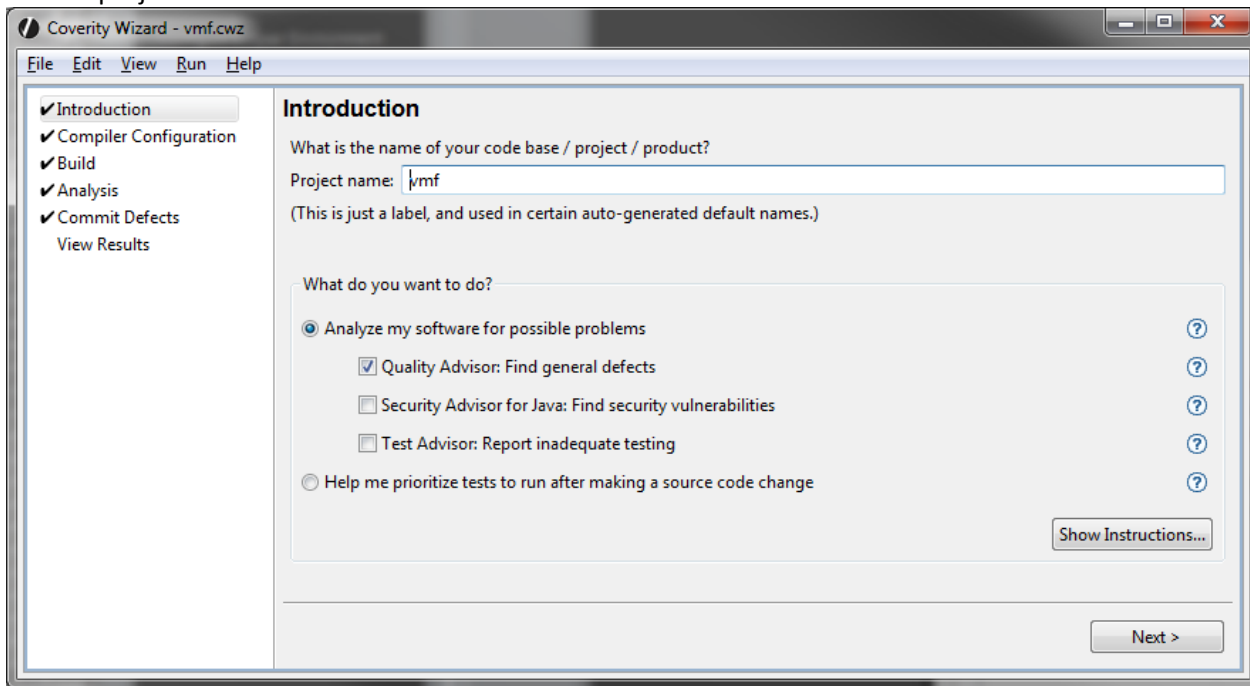
Menu → Help → Import New License File... → Browse for downloaded file → OK



Getting Started With Coverity Wizard

Introduction

Enter a project name.

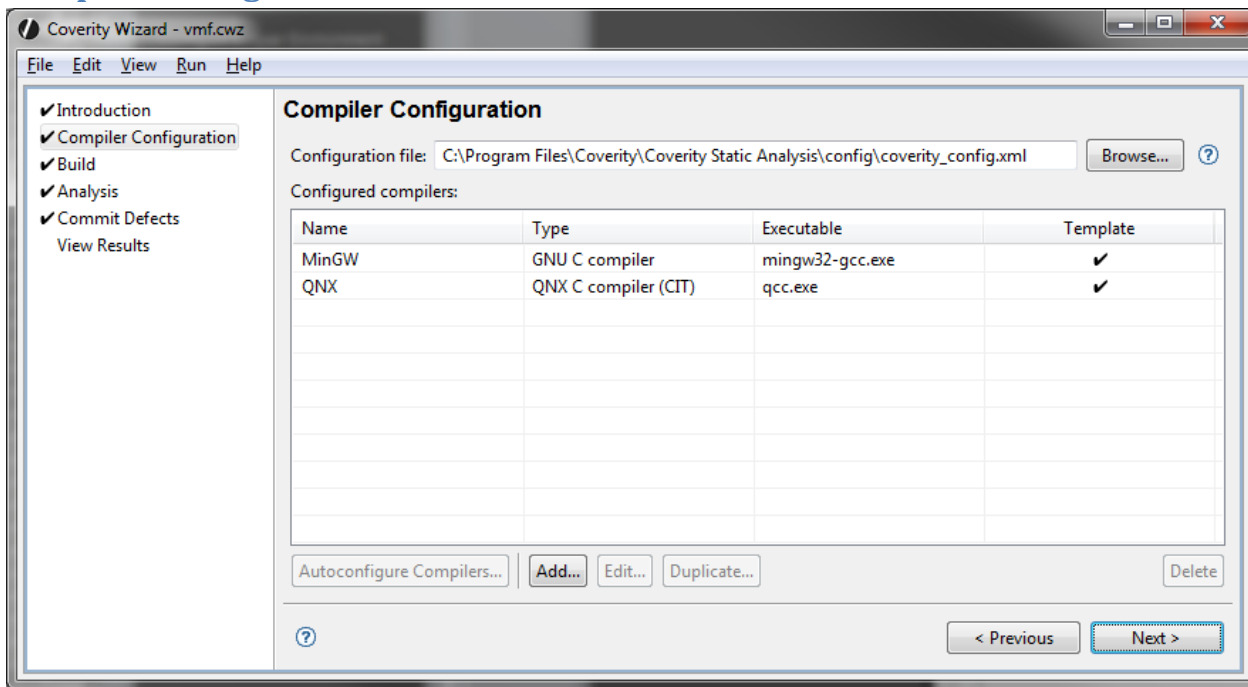


The screenshot shows the 'Coverity Wizard - vmf.cwz' window. On the left is a sidebar with a list of steps: Introduction (checked), Compiler Configuration (checked), Build (checked), Analysis (checked), Commit Defects (checked), and View Results. The main area is titled 'Introduction' and contains the following text: 'What is the name of your code base / project / product?'. Below this is a text input field with 'vmf' entered. A note below the field says '(This is just a label, and used in certain auto-generated default names.)'. Further down, under the heading 'What do you want to do?', there are four radio button options, each with a help icon (?):

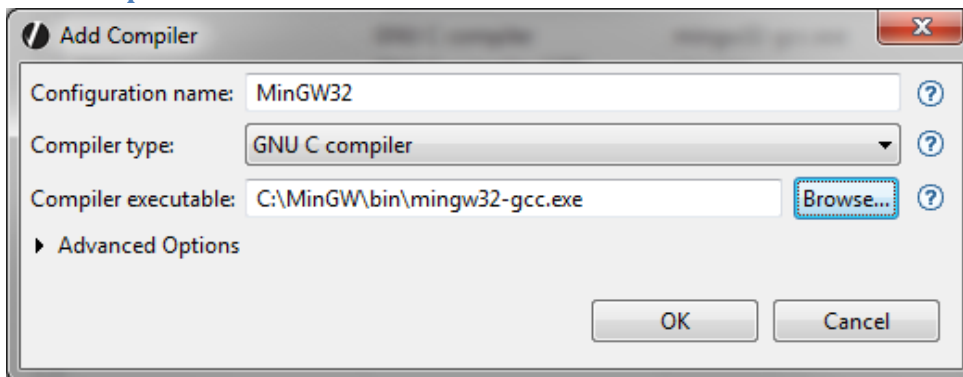
- ☒ Analyze my software for possible problems
 - ☒ Quality Advisor: Find general defects
 - ☐ Security Advisor for Java: Find security vulnerabilities
 - ☐ Test Advisor: Report inadequate testing
- ☐ Help me prioritize tests to run after making a source code change

At the bottom right of the main area is a 'Show Instructions...' button. At the very bottom of the window is a 'Next >' button.

Compiler Configuration



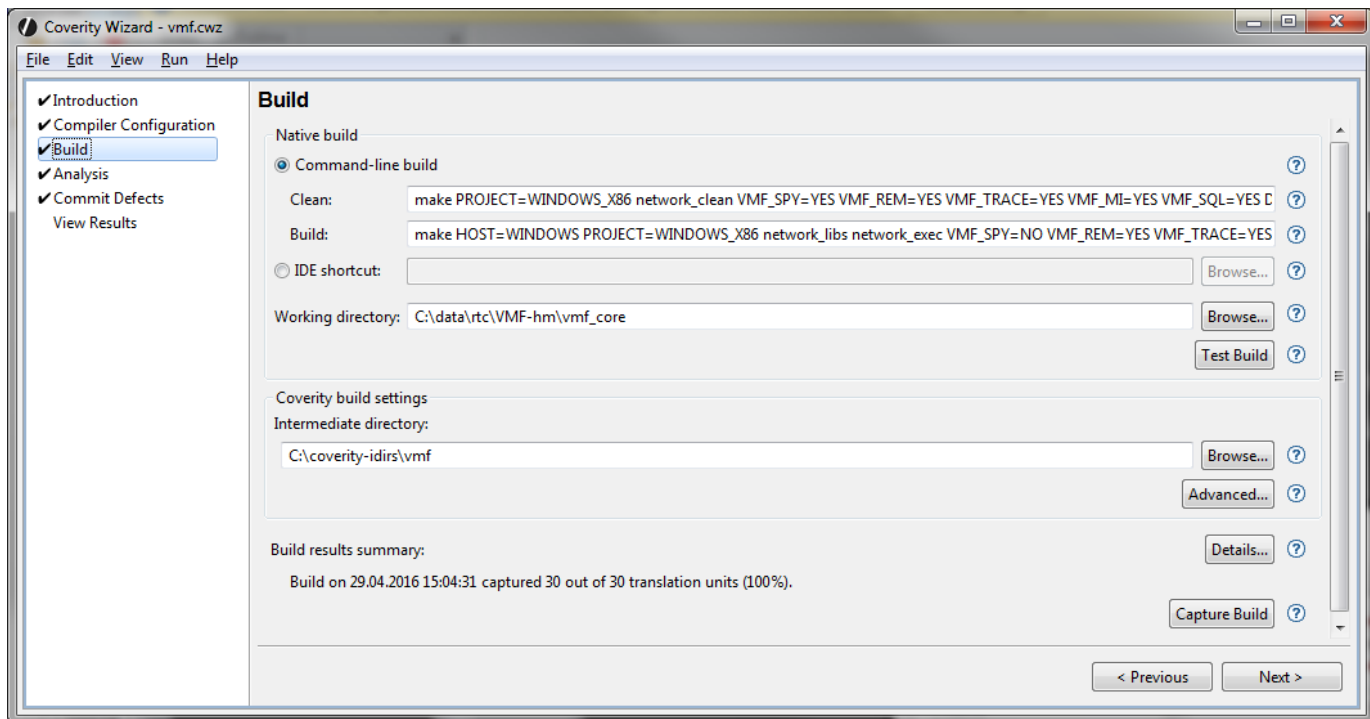
Add Compiler



1. Enter name for Compiler (e.g. MinGW32)
2. Browse for Compiler Executable

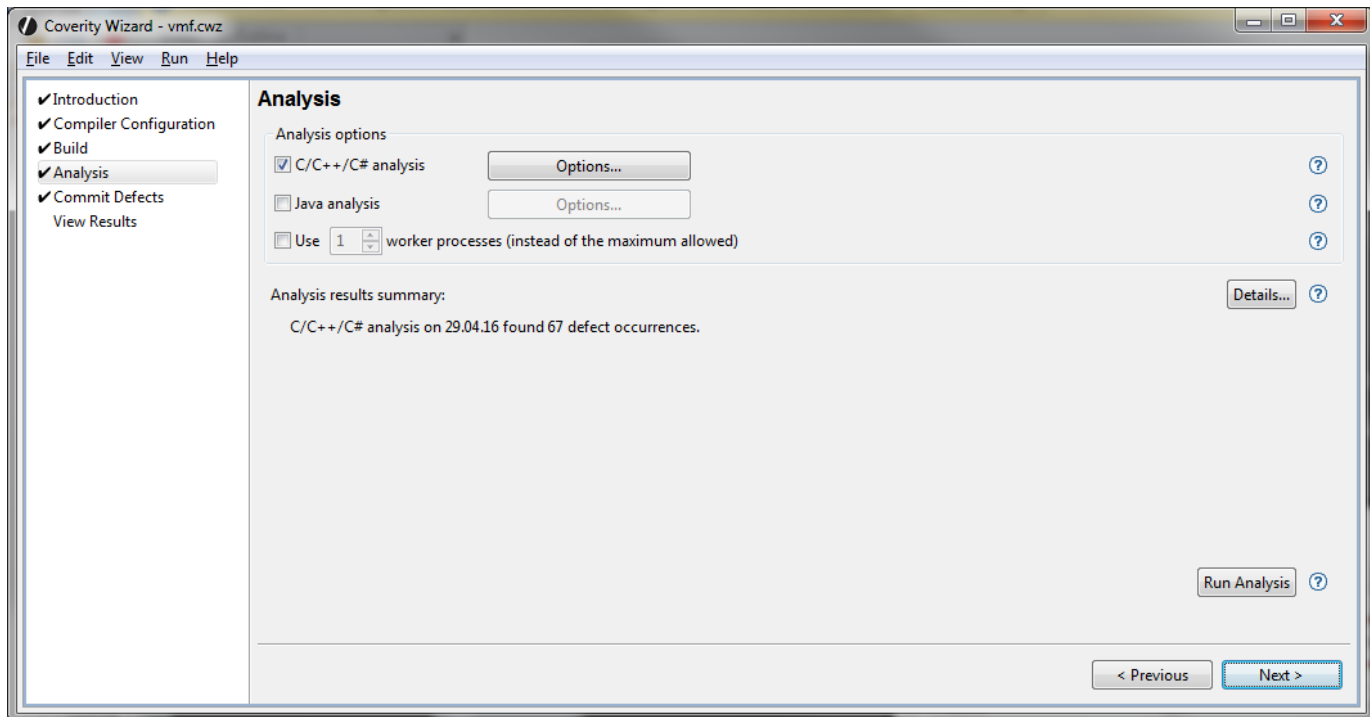
Note: For QNX Compiler select the qcc.exe (e.g. C:\qnx660\host\win32\x86\usr\bin\qcc.exe)!

Build



1. Enter the Clean and Build commands and working directory (where make shall be executed).
2. Check that build runs properly with "Test Build" button.
3. Enter intermediate directory.
4. Run "Capture Build"

Analysis



Press "Run Analysis".

Commit Defects

The screenshot shows the 'Commit Defects' window of the Coverity Wizard. The left sidebar has a tree view with the following items: Introduction, Compiler Configuration, Build, Analysis, Commit Defects (selected), and View Results. The main area is titled 'Commit Defects' and contains the following sections:

- Coverity Connect settings:**
 - Host name: chipd013.chennai.visteon.com
 - Port: 8080 ☐ Use SSL
 - Username: jmerkle
 - Password: ***** ☒ Save password
 - Test Connection**
- Streams to commit to:**
 - ☒ Commit C/C++/C# to stream: jmerkle_vmf **New...**
 - ☐ Commit Java to stream: **New...**
- Snapshot description:**
- Advanced Commit Settings...**
- Commit Defects**

At the bottom, there are buttons for '< Previous' and 'Next >'.

1. Enter credentials for Coverity Server
2. "Test Connection"
3. Select your stream which should appear in the dropdown
4. Press "Commit Defects"

Note: Server in Chennai allows users to create a Coverity Stream which is needed to push defects to the server. Check next section which explains how to create a Coverity Stream.

View Results

The screenshot shows the 'View Results' window of the Coverity Wizard. The left sidebar is the same as the previous window, with 'View Results' now selected. The main area is titled 'View Results' and contains the following text:

The source code and defects have been committed to Coverity® Connect. Browse to the following URL to see them:

http://chipd013.chennai.visteon.com:8080/query/defects.htm?stream=jmerkle_vmf&outstanding=true

At the bottom, there are buttons for '< Previous' and **Exit**.

With the link you should access directly the list of results in the browser.

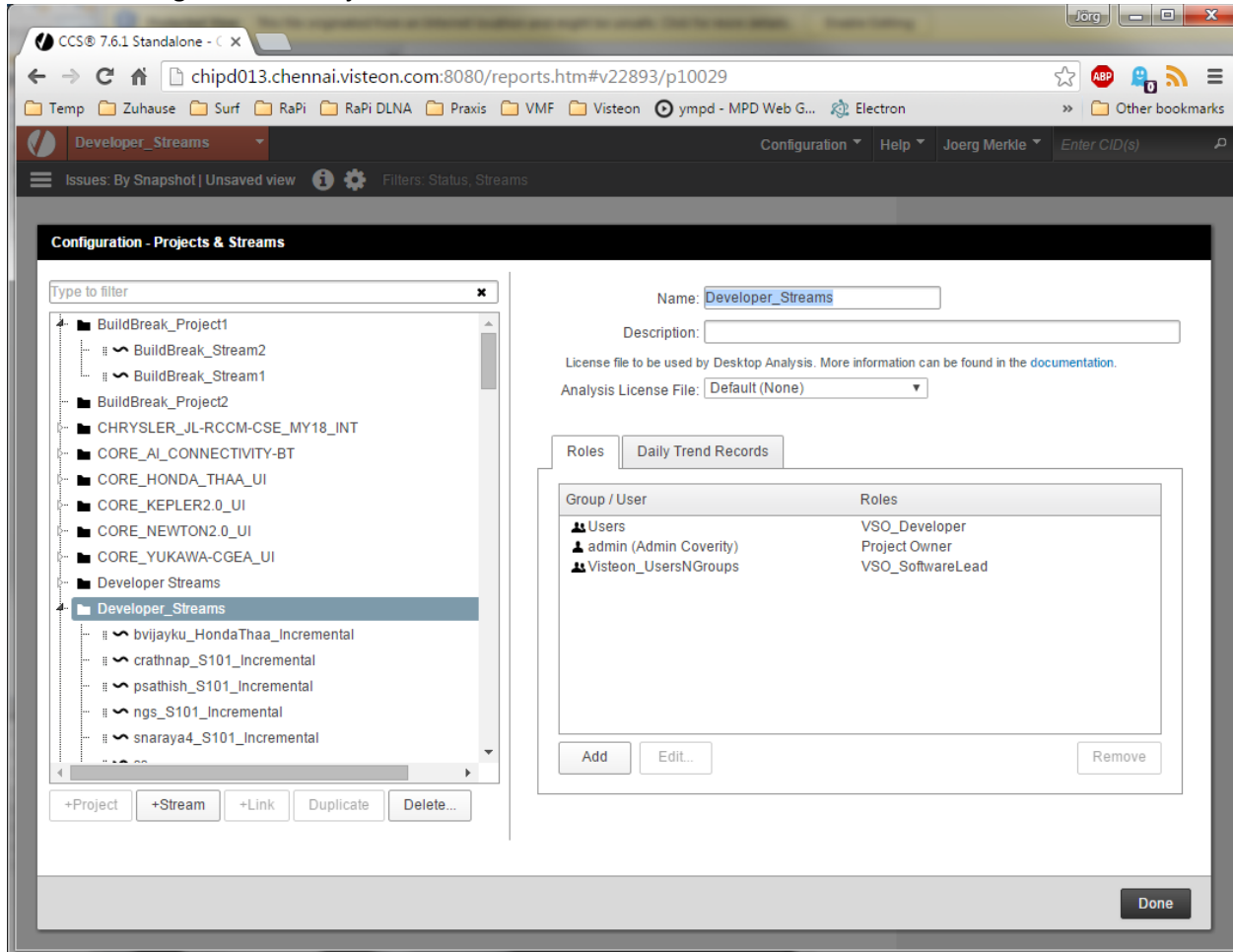
Coverity Connect

Coverity Connect is the server based part of Coverity which is needed to view and triage the defects. The local analysis doesn't have any viewer which means that after each run of the analysis the defects are uploaded to the server. Therefore a Coverity Stream needs to be available in order to commit the defects (don't mix up with RTC stream!).

Coverity Stream

Login in to the Chennai Coverity Website <http://chipd013.chennai.visteon.com>.

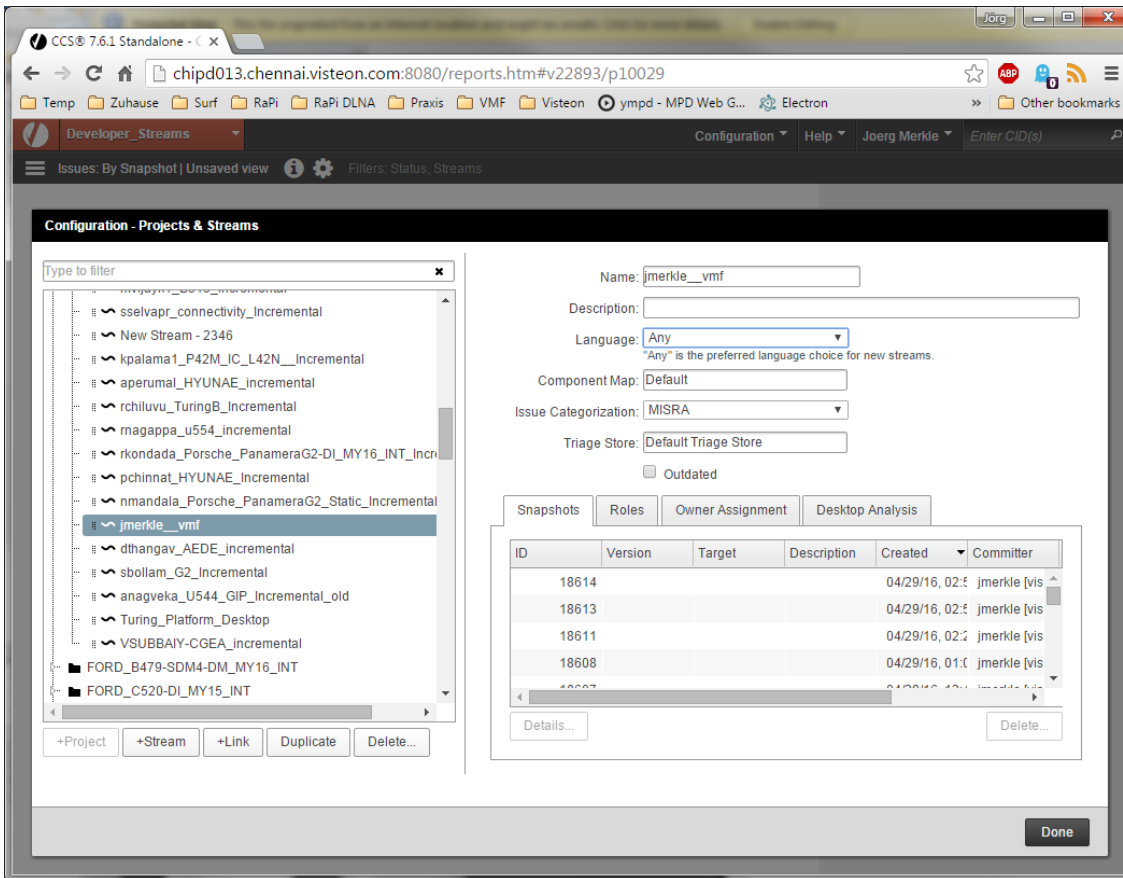
Menu → Configuration → Projects & Streams



1. Enter a name and press "Add" button.

The name of the stream shall then be available in the Coverity Wizard to commit defects.

Example configuration



Configuration - Projects & Streams

Type to filter

- sselvapr_connectivity_Incremental
- New Stream - 2346
- kpalama1_P42M_IC_L42N__Incremental
- aperumal_HYUNAE_incremental
- rchiluvu_TuringB_Incremental
- magappa_u554_incremental
- rkondada_Porsche_PanameraG2-DI_MY16_INT_Incr
- pchinna_HYUNAE_Incremental
- nmandala_Porsche_PanameraG2_Static_Incremental
- jmerkler_vmf**
- dthangav_AEDE_incremental
- sbollam_G2_Incremental
- anagveka_U544_GIP_Incremental_old
- Turing_Platform_Desktop
- VSUBBAIY-CGEA_incremental
- FORD_B479-SDM4-DM_MY16_INT
- FORD_C520-DI_MY15_INT

+Project +Stream +Link Duplicate Delete...

Name: jmerkler_vmf

Description:

Language: Any
"Any" is the preferred language choice for new streams.

Component Map: Default

Issue Categorization: MISRA

Triage Store: Default Triage Store

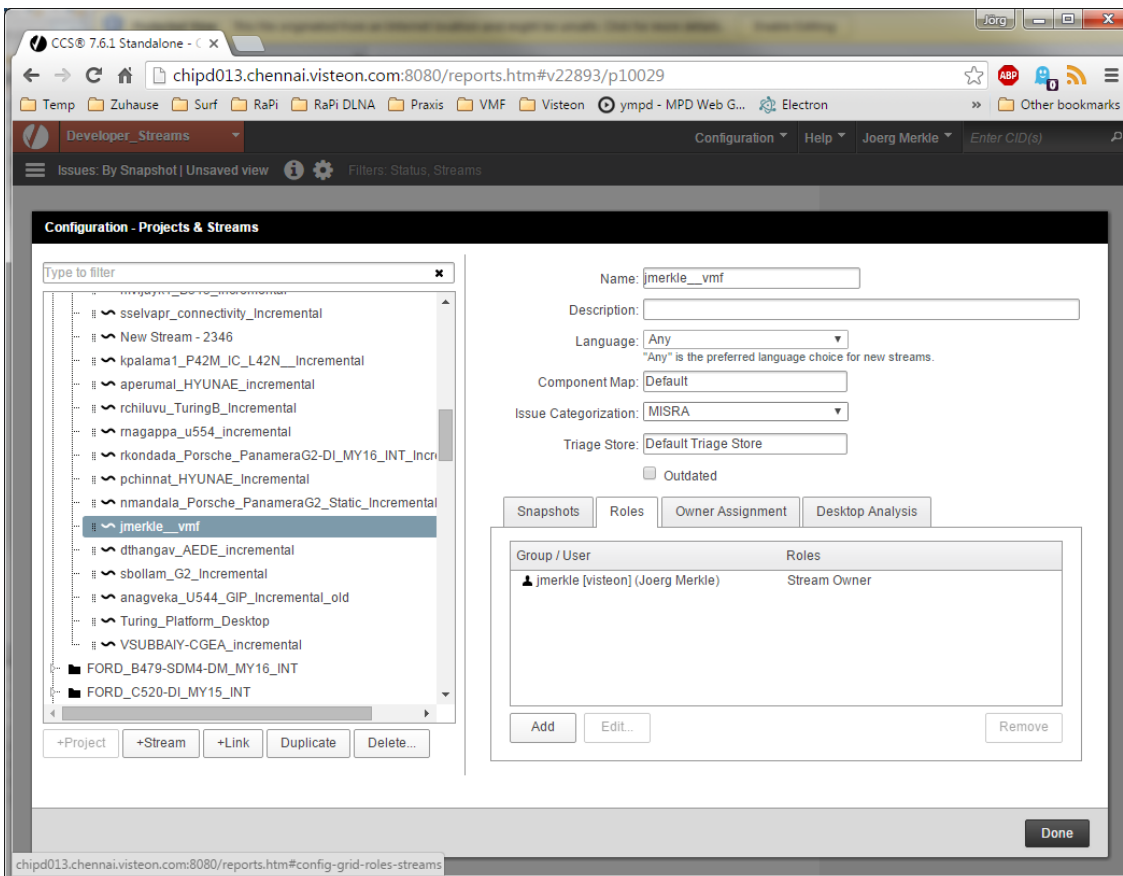
☐ Outdated

Snapshots Roles Owner Assignment Desktop Analysis

ID	Version	Target	Description	Created	Committer
18614				04/29/16, 02:11	jmerkler [vis
18613				04/29/16, 02:11	jmerkler [vis
18611				04/29/16, 02:11	jmerkler [vis
18608				04/29/16, 01:11	jmerkler [vis

Details... Delete...

Done



Configuration - Projects & Streams

Type to filter

- sselvapr_connectivity_Incremental
- New Stream - 2346
- kpalama1_P42M_IC_L42N__Incremental
- aperumal_HYUNAE_incremental
- rchiluvu_TuringB_Incremental
- magappa_u554_incremental
- rkondada_Porsche_PanameraG2-DI_MY16_INT_Incr
- pchinna_HYUNAE_Incremental
- nmandala_Porsche_PanameraG2_Static_Incremental
- jmerkler_vmf**
- dthangav_AEDE_incremental
- sbollam_G2_Incremental
- anagveka_U544_GIP_Incremental_old
- Turing_Platform_Desktop
- VSUBBAIY-CGEA_incremental
- FORD_B479-SDM4-DM_MY16_INT
- FORD_C520-DI_MY15_INT

+Project +Stream +Link Duplicate Delete...

Name: jmerkler_vmf

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☐ Outdated

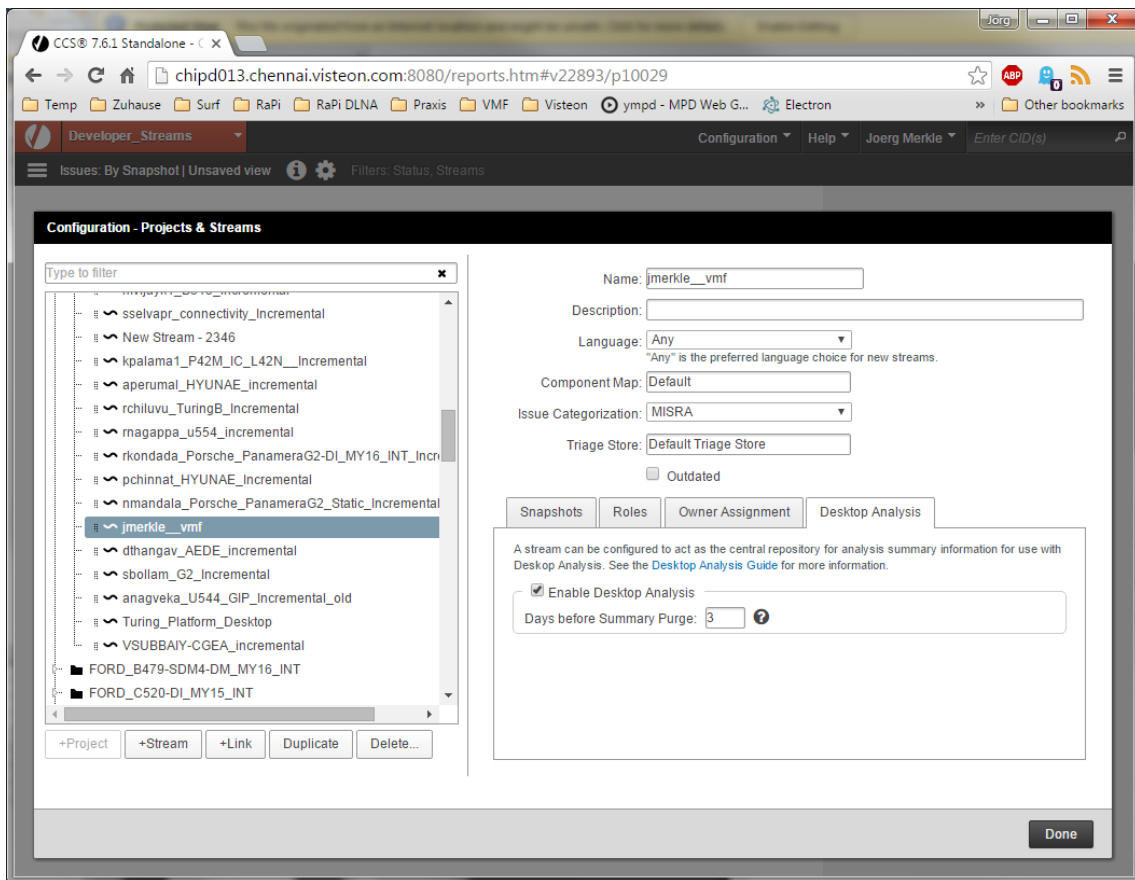
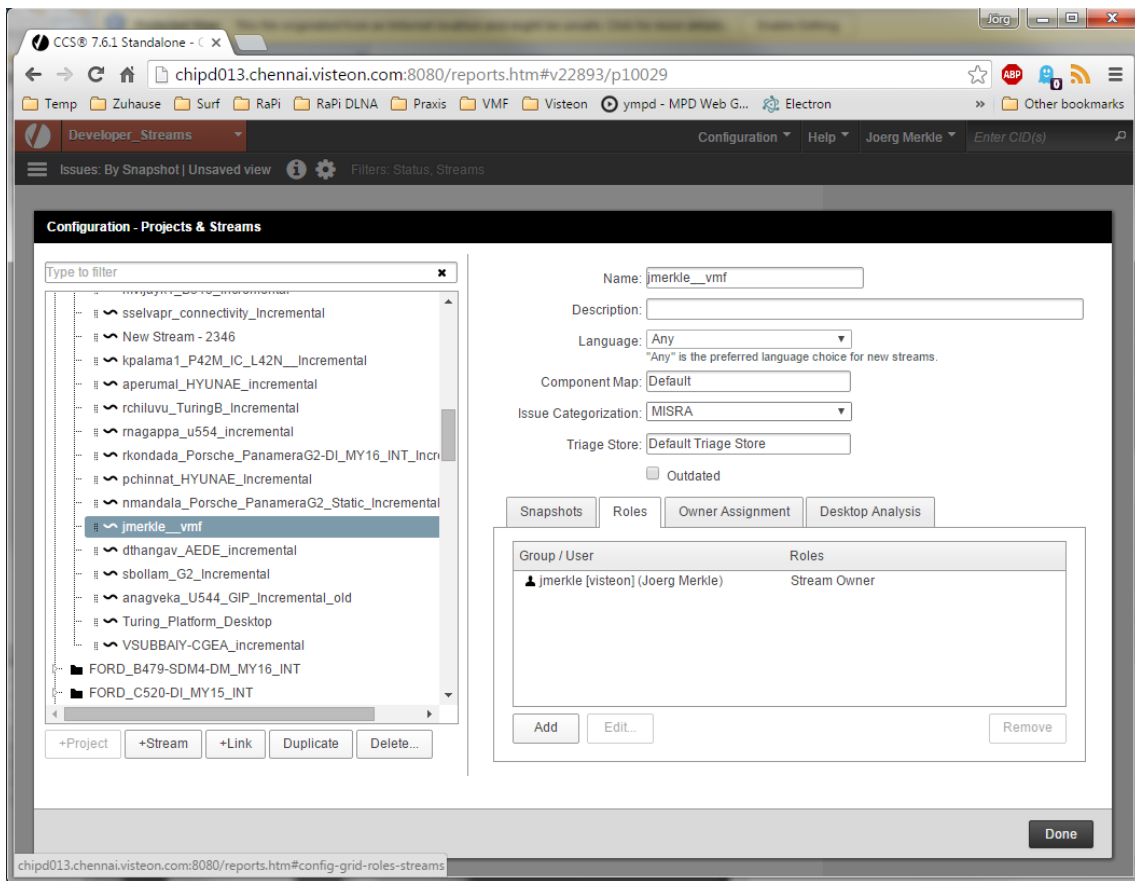
Snapshots Roles Owner Assignment Desktop Analysis

Group / User	Roles
jmerkler [visteon] (Joerg Merkle)	Stream Owner

Add Edit... Remove

Done

chipd013.chennai.visteon.com:8080/reports.htm#config-grid-roles-streams



VMF Coverity Setup

The Coverity Wizard allows to save its configuration. Unfortunately it doesn't seem to save the complete setup (e.g. for compilers).

The VMF configuration is located at `RTC vmf_core\coverity_cfg\vmf.cwz`.

The 2 batch files **coverity_win.bat** and **coverity_qnx.bat** are examples for the complete process of Coverity (Build, Analyse, Commit Defects) executed from command line (cmd.exe). These are the same steps as done in the Coverity Wizard. **In order to use it, you should first ensure the Coverity Wizard completes successfully!**